

- Supports SSH and X.509 for security, and even supports login token functionality.

Official website: <http://www.opennebula.org>

Latest version: 5.8.4

Eucalyptus

Eucalyptus is an acronym for Elastic Utility Computing Architecture for Linking Your Programs to Useful Systems. It is an open source software framework that provides the platform for private cloud computing implementation on computer clusters. Eucalyptus implements Infrastructure as a Service (IaaS) methodology for solutions in private and hybrid clouds.

Eucalyptus provides a platform for a single interface so that users can calculate the resources available in private clouds and the resources available externally in public cloud services. It is designed with extensible and modular architecture for Web services. It also implements the industry standard Amazon Web Services (AWS) API.

The Eucalyptus user console provides an interface for users to provision and configure compute, network and storage resources on their own. Eucalyptus can run multiple versions of Windows and Linux virtual machine images. Users can build a library of Eucalyptus machine images (EMIs) with application metadata that is decoupled from infrastructure details to allow them to run on Eucalyptus clouds.

Amazon Machine Images are also compatible with Eucalyptus clouds. VMware images and vApps can be converted to run on Eucalyptus clouds and AWS public clouds. Eucalyptus user identity management can be integrated with existing Microsoft Active Directory or LDAP systems to have fine-grained role-based access control over cloud resources. Eucalyptus supports storage area network devices to take advantage of storage arrays, thus improving

performance and reliability. Eucalyptus machine images can be backed by EBS-like persistent storage volumes, improving the performance of image launch time and enabling fully persistent virtual machine instances. Eucalyptus also supports direct-attached storage.

Architecture: The Eucalyptus architecture has the following five main components.

- **Cloud controller (CLC):** CLC acts as the administrative interface for cloud management and performs high-level resource scheduling and system accounting. The CLC accepts user API requests from command-line interfaces like euca2ools or GUI-based tools like the Eucalyptus management console, and manages the underlying computer storage and network resources.
- **Scalable object storage (SOS):** This is a pluggable service that allows infrastructure administrators the flexibility to implement scale-out storage on top of commodity resources using open source and commercial solutions that implement the S3 interface.
- **Cluster controller (CC):** Written in C, this acts as the front-end for clusters within the Eucalyptus cloud and communicates with the storage and node controllers.

- **Storage controller (SC):** Written in Java, this communicates with the cluster controller and the node controller, managing Eucalyptus block volumes and snapshots to instances within its specific cluster. It interfaces with storage systems including Local, NFS, iSCSI and SAN.
- **Node controller (NC):** This is written in C, hosts the virtual machine instances and manages the virtual network endpoints. It caches images from scalable object storage, and creates and caches instances.

Features

- Works with multiple hypervisors including VMware, Xen and KVM.
- Communication within internal processes is secured through SOAP and WS-Security.
- Offers administrative features such as user and group management, and reports.
- Well-defined interfaces (via WSDL, since they are Web services) and thus can be easily swapped out for custom components.
- Flexible and can be installed on a very minimal setup.

Official website: <https://www.eucalyptus.cloud/>

Latest version: 4.4.3 

References

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- [5] <http://www.mist.io>
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By: Anand Nayyar

The author works at Duy Tan University in Vietnam. He loves to explore open source technologies, sensor communications, network security and the Internet of Things.